Chapter 12: Project Procurement Management

Information Technology Project Management, Ninth Edition

Note: See the text itself for full citations

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Learning Objectives (1 of 2)

- Explain the importance of project procurement management and the increasing use of outsourcing for information technology (IT) projects
- Describe the work involved in planning procurements for projects, including determining the proper type of contract to use, performing a make-or-buy analysis, and preparing a procurement management plan, statement of work, and source selection criteria
- Discuss how to conduct procurements and strategies for obtaining seller responses, selecting sellers, and awarding contracts
- Describe the process of controlling procurements by managing procurement relationships and monitoring contract performance

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Learning Objectives (2 of 2)

- Discuss types of software that are available to assist in project procurement management
- Discuss considerations for agile/adaptive environments

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Importance of Project Procurement Management

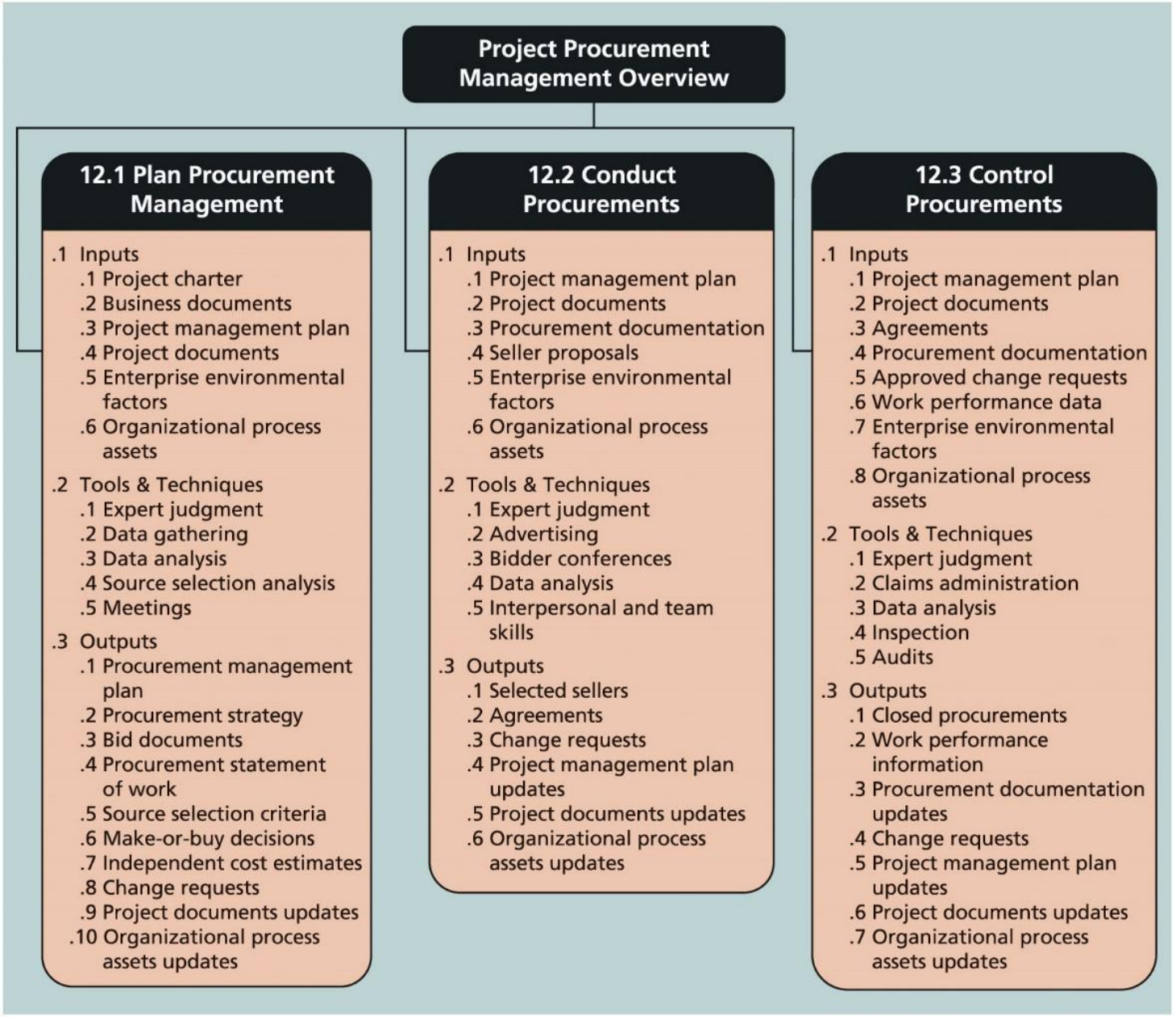
- Procurement means acquiring goods and/or services from an outside source
  - Other terms include purchasing and outsourcing
  - PMI defines outside source as source outside the project team, i.e. the same organization can be a supplier to the project team
- Why outsource?
  - Access skills and technologies
  - Allow the client organization to focus on its core business
  - Reduce both fixed and recurrent costs
    - Labor costs of hiring, firing, reassigning people to projects, or paying their salaries when they are between projects
  - Provide flexibility
    - Extra staff during peak workloads
  - Increase accountability
    - Contract are legally binding and there is more accountability for delivering work as stated in the contract

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Project Procurement Management

- Main processes
  1. Planning procurement management
     - Determining what to procure and when and how to do it
  2. Conducting procurements
     - Obtaining seller responses, selecting sellers, and awarding contracts
  3. Controlling procurements
     - Managing relationships with sellers, monitoring contract performance, making changes as needed, and closing out contracts

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FIGURE 12-1 Project procurement management overview

1. Planning Procurement Management

- Identifying which project needs can best be met by using products or services outside the organization
  - Involves deciding whether to procure, what to procure, how to procure, how much to procure, and when to procure
  - An important output of this process is the make-or-buy decision
  - If there is no need to buy any products or services from outside the organization, then there is no need to perform any of the other procurement management processes

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Types of Contracts (1 of 7)

- Fixed price or lump sum contracts : involve a fixed total price for a well-defined product or service
  - Firm-fixed-price (FFP)
  - Fixed-price with economic price adjustment (FP-EPA)
    - Include provision for pre-defined final adjustment to the contract price due to market conditions that are beyond the control of both the buyer and seller, e.g. changes in the cost of labor and material, general inflation, fluctuations in the currency market
    - Beneficial for long-term projects or those that span multiple years

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Types of Contracts (2 of 7)

- Fixed price or lump sum contracts (cont.)
  - Fixed-price incentive fee (FPIF)
    - Point of Total Assumption (PTA): the total cost above which the seller bears all the costs of a cost overrun. Any cost overrun above the PTA is not shared by the buyer and totally absorbed by the seller

$$PTA = ((Ceiling\ Price - Target\ Price) / Buyer's\ Share\ Ratio) + Target\ Cost$$

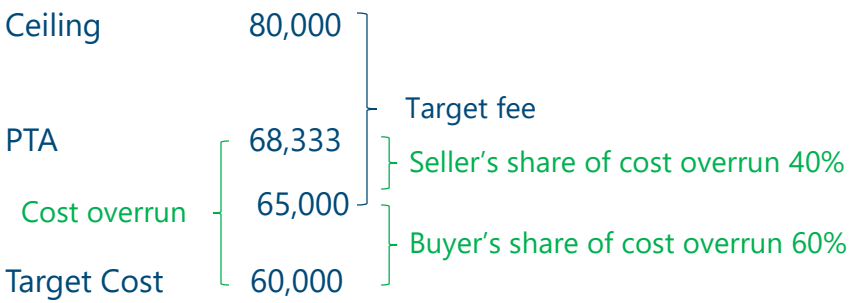
Term	Description
Target Cost	The estimated cost of the project. It does not include seller's profit
Target Fee	Seller's Fee or Profit
Target Price	Total estimated price of the project (including seller's fee). Target Price = Target Cost + Target Fee
Buyer's Share Ratio (BSR)	In case of cost overrun, the ratio of the cost that buyer will bear.
Ceiling Price	Maximum Price that the buyer is willing to pay - really the maximum, not even a penny more! It includes Target Price plus buyer's share of the cost overrun.

https://www.deepfriedbrainproject.com/2009/09/point-of-total-assumption-pta-fpif-pmp.html

Types of Contracts (3 of 7)

- Fixed price or lump sum contracts (cont.)
  - Fixed-price incentive fee (FPIF) (cont.)
    - Example terms of the contract

```
Target Cost: $60,000
Target Fee: $15,000
Target Price: $75,000
Ceiling Price: $80,000
Buyer-Seller Share Ratio: 60:40
```



- PTA = ((80,000 - 75,000)/0.6) + 60,000 = (5,000/0.6) + 60,000 = 8,333 + 60,000 = $68,333
- Maximum cost overrun that buyer will share cost = 68,333 - 60,000 = $8,333
- Buyer's share of cost overrun = 0.6*8,333 = $5,000
- Amount buyer pays at PTA = Target Price (or Target Cost + Target Fee) + Buyer's share of cost overrun = 75,000 + 5,000 = $80,000 (= Ceiling Price)
- The amount seller received at PTA = the amount buyer pays = $80,000
- The amount seller spent at PTA (actual cost) = $68,333
- The amount seller makes at PTA = 80,000 - 68,333 = + $11,666 (profit)
- In this case, seller is still making some profit even when the actual cost reaches PTA

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https://www.deepfriedbrainproject.com/2009/10/point-of-total-assumption-pta-facts.html

Types of Contracts (4 of 7)

- Cost-reimbursable contracts
  - Seller is reimbursed the actual costs that incur in carrying out the works, plus an additional fee
  - Cost plus incentive fee (CPIF)
    - The buyer pays the seller for allowable costs (as defined in the contract) plus a predetermined fee and incentive bonus
    - Incentives are provided for completing the project ahead of schedule or for reducing contract costs
  - Cost plus fixed fee (CPFF)
    - The buyer pays the seller for allowable costs (as defined in the contract) plus a fixed fee usually based on a percentage of estimated costs
  - Cost plus award fee (CPAF)
    - The buyer pays the seller for allowable costs (as defined in the contract) plus an award fee based on buyer's satisfaction
  - Cost plus percentage of costs (CPPC)
    - The buyer pays the seller for allowable costs (as defined in the contract) plus a predetermined percentage based on total costs
    - Least desirable type of contract because the seller has no incentive to decrease costs

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Types of Contracts (5 of 7)

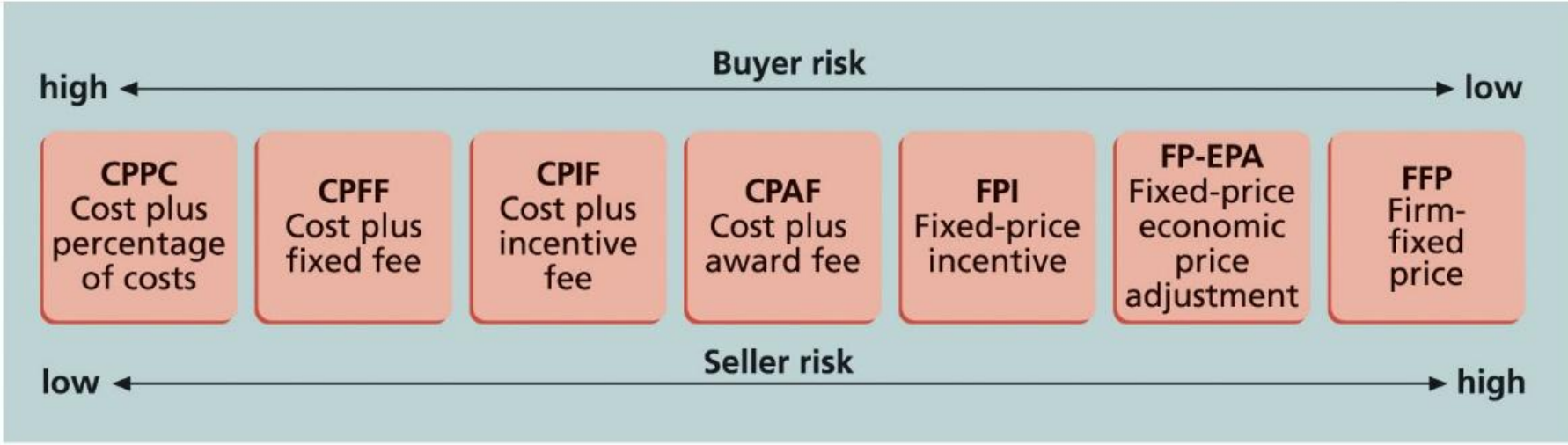


FIGURE 12-2 Contract types versus risk

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Types of Contracts (6 of 7)

- Time and material contracts
  - Hybrid of both fixed price and cost reimbursable contracts
  - Based on actual time spent (e.g. hourly or daily rates), and actual materials used (e.g. hardware, software licenses, equipment, etc.).
  - E.g. a fee of $80 per hour for services, plus a fixed price of $10,000 for providing a specific project material (or a ceiling of $10,000)
  - Might include "not-to-exceed" clause in a contract
  - Often used when total costs cannot be specified in a contract
  - Preferred by many contract programmers and consultants
- Unit price contracts
  - Require the buyer to pay the seller a predetermined amount per unit of service or material.
  - E.g. the value of contract is $2,500 for laying wire $25/$m^2$ for 100 m^2 or $10,000 for buying 10 units of $1,000 computer
  - Often involve volume discount

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Types of Contracts (7 of 7)

- Contracts should include specific clauses to take into account issues unique to the project, e.g.
  - Time and material contracts should stipulate different hourly rates based on the level of experience of individual contractors
  - A termination clause allows the buyer or supplier to end the contract

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Tools and Techniques for Planning Procurement Management (1 of 2)

- Make-or-buy analysis
  - Determine whether an organization should
    - Make or perform a particular product or service inside the organization vs. buy from someone else, or
    - Purchase vs. lease items for the project
    - E.g. Equipment has a purchase price of $12,000 and daily operating costs of $400 vs. lease for $800 per day
      - Determine when the purchase cost equals the lease cost:

        $800/day = $12,000 + $400/day

        $400/day = $12,000

        d = 30
      - If equipment is needed < 30 days, lease it
      - If equipment is needed > 30 days, purchase it

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Tools and Techniques for Planning Procurement Management (2 of 2)

- Expert judgment
  - Experts both from inside and outside an organization can provide excellent advice in planning purchases and acquisitions
- Market research
  - Many potential suppliers are often available for goods and services
  - Information is available online and conferences are held where attendees can see and discuss new products

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Procurement Management Plan

- Describes how the procurement processes will be managed, from developing documentation for making outside purchases or acquisitions to contract closure
  - Guidelines for types of contracts to be used
  - Standard procurement documents or templates
  - Guidelines for creating contract WBS, statements of work, and other documents
  - Roles and responsibilities of project team and related departments, such as purchasing and legal department
  - Guidelines for using independent estimates to evaluate sellers
  - Lead times for purchase and acquisition
  - Guidelines for identifying prequalified sellers and organizational lists of preferred sellers
  - Etc.

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Statement of Work

- A contract SOW is a type of scope statement that describes the work required for a particular contract
- A good contract SOW gives bidders a better understanding of the buyer's expectations
  - Becomes part of the official contract

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Statement of Work (SOW)

I. **Scope of Work:** Describe the work to be done in detail. Specify the hardware and software involved and the exact nature of the work.

II. **Location of Work:** Describe where the work must be performed. Specify the location of hardware and software and where the people must perform the work.

III. **Period of Performance:** Specify when the work is expected to start and end, working hours, number of hours that can be billed per week, where the work must be performed, and related schedule information.

IV. **Deliverables Schedule:** List specific deliverables, describe them in detail, and specify when they are due.

V. **Applicable Standards:** Specify any company or industry-specific standards that are relevant to performing the work.

VI. **Acceptance Criteria:** Describe how the buyer organization will determine if the work is acceptable.

VII. **Special Requirements:** Specify any special requirements such as hardware or software certifications, minimum degree or experience level of personnel, travel requirements, and so on.

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FIGURE 12-3 Statement of work (SOW) template

Procurement or Bid Documents

- Request for Proposals (RFP)
  - Used to solicit proposals from prospective sellers
  - A proposal is a document prepared by a seller when there are different approaches for meeting buyer needs
- Request for Quotes (RFQ)
  - Used to solicit quotes or bids from prospective suppliers
  - A bid, also called a tender or quote (short for quotation), is a document prepared by sellers providing pricing for standard items that have been clearly defined by the buyer

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Source Selection Criteria

- It's important to prepare some form of evaluation criteria, preferably before issuing a formal RFP or RFQ
- Organizations use criteria to rate or score proposals, and they often assign a weight to each criterion to indicate its importance, e.g.
  - Technical approach 30%
  - Management approach 30%
  - Past performance 20%
  - Price 20%
- Some IT projects also require potential sellers to deliver a technical presentation as part of their proposal
  - The proposed project manager should lead the potential seller's presentation team

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Examples from a Real RFP

- It's very important for buyers to clarify what they want so that sellers can respond appropriately
- See attachment document 
  - Example of Houstonfirst's RFP for project management services (to assist rebuilding after flooding from Hurricane Harvey)
    - https://www.houstonfirst.com/do-business/
  - Letter of clarification from RFP

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2. Conducting Procurements (1 of 4)

- Steps after planning for procurement management
  - Decide whom to ask to do the work
  - Send appropriate documentation to potential sellers
  - Obtain proposals or bids
  - Select a seller
  - Award a contract

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Conducting Procurements (2 of 4)

- Approaches for procurement
  - Organizations can advertise to procure goods and services in several ways
    - Approaching the preferred supplier
    - Approaching several potential suppliers and taking bids
- A bidders' conference can help clarify expectations
  - Meeting with prospective sellers prior to preparation of their proposals or bids
- A technical team, a management team, and a cost team evaluate proposals or bids from sellers
  - Use a proposal evaluation sheet to create a short list of best 3-5 proposals

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Conducting Procurements (3 of 4)

		Proposal 1		Proposal 2		Proposal 3		Etc.	
Criteria	**Weight**	**Rating**	**Score**	**Rating**	**Score**	**Rating**	**Score**		
Technical approach	30%	90	27	80	24	70	21		
Management approach	30%	85	25.5	75	22.5	85	25.5		
Past performance	20%	95	19	70	14	75	15		
Price	20%	75	15	95	19	80	16		
Total score	**100%**		**86.5**		**79.5**		**77.5**		

FIGURE 12-4 Sample proposal evaluation sheet

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Conducting Procurements (4 of 4)

- Seller selection
  - After developing a short list of possible sellers, organizations often follow a more detailed proposal evaluation process
  - It is customary to have contract negotiations during the source selection process
  - Sellers on the short list often prepare a best and final offer (BAFO)
  - Final output is a contract signed by the buyer and the selected seller

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3. Controlling Procurements (1 of 4)

- Ensures the seller's performance meets contractual requirements
  - Contracts are legal relationships, so it is important that legal and contracting professionals be involved in writing and administering contracts
    - Project team members must be aware of potential legal problems they might cause by not understanding a contract
- It is critical that project managers and team members watch for constructive change orders
  - A situation when a contractor performs work beyond the contract requirements, without a formal order under the changes clause, either due to an informal order from, or through the fault of, the buyer
  - Contractor may be entitled to payment for additional costs incurred, and an extension to the contract performance period.
  - This is also a change where the buyer and the seller cannot reach an agreement and could result in claims/disputes

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Controlling Procurements (2 of 4)

- Suggestions for change control in contracts
  - Use tools and techniques, such as a contract change control system, buyer-conducted performance reviews, inspections and audits, etc.
  - Changes to any part of the project need to be reviewed, approved, and documented by the same people in the same way that the original part of the plan was approved
  - Evaluation of any change should include an impact analysis
  - Changes must be documented in writing
  - Project managers and teams should stay closely involved to make sure the new system will meet business needs and work in an operational environment
  - Have backup plans in case the new system does not work as planned

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Controlling Procurements (3 of 4)

- The contract itself should include requirements for formal acceptance and closure
- Closing procurements
  - Involves completing and settling contracts and resolving any open items
  - The project team should determine if all work was completed correctly and satisfactorily, update records to reflect final results, and archive information for future use

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Controlling Procurements (4 of 4)

- Tools to assist in contract closure
  - Procurement audits identify lessons learned in the procurement process
  - A records management system provides the ability to easily organize, find, and archive procurement-related documents for future use
  - Ideally, all procurements should end in a negotiated settlement between the buyer and the seller
    - If negotiation is not possible, then some type of alternative dispute resolution (ADR) such as mediation or arbitration can be used

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Using Software to Assist in Project Procurement Management

- Word-processing software: write proposals and contracts
- Spreadsheet software: create proposal evaluation worksheets
- Databases: track suppliers
- Presentation software: present procurement-related information
- E-procurement software: electronic procurement functions

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Considerations for Agile/Adaptive Environments

- Agile Manifesto values customer collaboration over contract negotiation, setting an important tone for procurement relationships on agile projects
  - The buyer and seller should work together to create the required products and services throughout the entire procurement process
- Another goal of agile/adaptive environments is speed
  - Several procurements, however, take time
  - Plan for procurement needs in advance

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Chapter Summary

- Procurement, purchasing, or outsourcing is the acquisition of goods and services from an outside source
  - Project procurement management includes planning procurement management and then conducting and controlling procurements
    - Involves deciding what to procure or outsource, what type of contract to use, and how to describe the effort in a statement of work
  - Several types of software can assist in project procurement management

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